Joao A Ascensao

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EDUCATION

2017–2023 **University of California, Berkeley** · *Ph.D. in Bioengineering*
Concentration in Biophysics & Computational Biology · Advisor: Oskar Hallatschek

2012–2016 **Rice University** · *B.S. in Bioengineering*
Minor in Global Health Technologies · Advisor: Oleg Igoshin

ACADEMIC POSITIONS

2024–present **Harvard University** — *Cambridge, MA*
Postdoctoral Fellow · Advisor: Michael Desai

2016–2017 **Universitat Pompeu Fabra** — *Barcelona, Spain*
Fulbright Scholar & Whitaker Fellow · Advisor: Jordi Garcia-Ojalvo

Summer 2015 **University of California, Berkeley** — *Berkeley, CA*
Amgen Scholar · Advisor: Adam Arkin

Summer 2014 **Rice Global Health & St. Gabriel's Hospital** — *Namitete, Malawi*
Biomedical Devices Intern

Summer 2013 **The Jackson Laboratory** — *Bar Harbor, ME*
NSF-REU Intern · Advisor: Judith Blake

PUBLICATIONS & PREPRINTS

1. **Ascensao JA**, Yu Q, Hallatschek O. The evolution of genetic drift over 50,000 generations. *Genetics* (2026). doi: 10.1093/genetics/iyag199
2. **Ascensao JA**, Abedi K, Prasad A, Hallatschek O. Frequency-dependent fitness effects are ubiquitous in rapidly adapting microbial populations. *Science Advances* (2026). doi: 10.1126/sciadv.aeh3295
3. **Ascensao JA**, Desai M. Experimental evolution in an era of molecular tools for manipulating genotypes and phenotypes. *Nature Reviews Genetics* (2026). doi: 10.1038/s41576-025-00867-6
4. **Ascensao JA**, Lok K, Hallatschek O. Asynchronous abundance fluctuations can drive giant genotype frequency fluctuations. *Nature Ecology and Evolution* (2025). doi: 10.1038/s41559-024-02578-3
5. **Ascensao JA**, Denk J, Lok K, Yu Q, Wetmore KM, Hallatschek O. Rediversification following ecotype isolation reveals hidden adaptive potential. *Current Biology* (2024). doi: 10.1016/j.cub.2024.01.029
   – See dispatch article by W. Shoemaker
6. Yu Q, **Ascensao JA**, Okada T, COG-UK consortium, Boyd O, Volz E, Hallatschek O. Lineage frequency time series reveal elevated levels of genetic drift in SARS-CoV-2 transmission in England. *PLOS Pathogens* (2024). doi: 10.1371/journal.ppat.1012090
7. **Ascensao JA**, Wetmore KM, Good BH, Arkin AP, Hallatschek O. Quantifying the local adaptive landscape of a nascent bacterial community. *Nature Communications* (2023). doi: 10.1038/s41467-022-35677-5
8. **Ascensao JA***, Datta P*, Hancioglu B*, Sontag E*, Gennaro ML, Igoshin OA. Non-monotonic response to monotonic stimulus: regulation of glyoxylate shunt gene expression dynamics in *Mycobacterium tuberculosis*. *PLOS Computational Biology* (2016). doi: 10.1371/journal.pcbi.1004741
9. **Ascensao JA***, Dolan ME*, Hill DP, Blake JA. Methodology for the inference of gene function from phenotype data. *BMC Bioinformatics* (2014). doi: 10.1186/s12859-014-0405-z *equal contribution

AWARDS & RECOGNITIONS

2025 CFAR Scholar Award
2022 UC Berkeley International Conference Travel Grant
2020 Lloyd Scholarship *(departmental award)*
2019 Brodie Scholarship *(departmental award)*
2017–22 NSF Graduate Research Fellowship
2017–22 Berkeley Fellowship for Graduate Study
2016–17 Fulbright Scholar
2016–17 Whitaker International Program Fellowship
2016 MIT Sloan Scholar for Graduate Study *[declined]*
2016 Princeton Presidential Fellowship & Dean's Grant *[declined]*
2016 Rice Institute for Global Health–Innovator Award
2015 Outstanding Junior in Bioengineering Award
2015 Best Engineering Poster; Rice Undergraduate Research Symposium
2015 Amgen Scholar; University of California, Berkeley
2014 Global Health Award; Rice Undergraduate Research Symposium
2012–16 President's Honor Roll

TALKS

2026	Physics of Living Systems Seminar, MIT	*Cambridge, MA*
2026	Condensed Matter Theory Seminar, Harvard University	*Cambridge, MA*
2024	Society of Industrial & Applied Math, Life Sciences Conference	*Portland, OR*
2023	Molecular Mechanisms in Evolution, Gordon Conference	*Easton, MA*
2023	Seminar at Max Planck Institute for the Science of Light	*Erlangen, Germany*
2022	Center for Theoretical Evolutionary Genomics	*Berkeley, CA*
2022	Bay Area Population Genetics	*Berkeley, CA*
2022	Evolutionary Dynamics and Processes	*Plön, Germany*
2020	Microbial Ecology and Evolution Virtual (MEEVirtual)	*Virtual*
2019	West Coast Bacterial Physiology	*Asilomar, CA*
2019	UC Berkeley Bioengineering Retreat (Brodie Scholar Talk)	*Santa Cruz, CA*
2017	Whitaker Fellows Seminar	*Lisbon, Portugal*
2013	Gene Ontology Consortium Meeting	*Bar Harbor, ME*

POSTERS

2026	Function of Evolving Systems, Gordon Conference	*Waterville Valley, NH*
2025	Microbial Population Biology, Gordon Conference	*Andover, NH*
2024	The Allied Genetic Conference (TAGC)	*National Harbor, MD*
2023	Bay Area Population Genetics	*Palo Alto, CA*
2023	Les Houches Theoretical Biological Physics	*Les Houches, France*
2023	Molecular Mechanisms in Evolution, Gordon Conference	*Easton, MA*
2023	Stochastic Physics in Biology, Gordon Conference	*Ventura, CA*
2023	Berkeley Statistical Mechanics Meeting	*Berkeley, CA*
2019	Microbial Population Biology, Gordon Conference	*Andover, NH*
2015	Bioengineering Undergraduate Poster Symposium	*Houston, TX*
2015	UC Berkeley Amgen Scholars Poster Session	*Berkeley, CA*
2015	Rice Undergraduate Research Symposium	*Houston, TX*
2014	Bioengineering Undergraduate Poster Symposium	*Houston, TX*

RESEARCH SUPERVISION

2024–present	Alex Poret, graduate student, biophysics
2024–present	Jenya Belousova, graduate student, evolutionary biology
2023–2025	Keon Abedi, undergraduate student, physics
spring 2024	Kira Buttrey, rotation student, bioengineering
fall 2023	Sarah Wasinger, rotation student, bioengineering
fall 2023	Aaron Fultineer, rotation student, physics
2020–2023	Kristen Lok, undergraduate student, bioengineering
2020–2021	Can Goksal, undergraduate student (co-supervised with QinQin Yu), biology
2019–2020	Nicole Tin, undergraduate student, bioengineering

GRADUATE COURSEWORK

2017	Statistical Mechanics
2017	Computational Molecular Biology
2018	Computational Statistics
2018	Genomics for Microbiology
2018	Mathematical Models of Infectious Disease
2018	Microbial Diversity & Evolution
2018	Biological Regulatory Mechanisms
2019	Non-equilibrium Statistical Mechanics
2019	Probabilistic Models & Machine Learning in Genomics
2024	Immunology

ADDITIONAL EDUCATION

Summer 2019	Microbial Diversity	*Marine Biological Laboratory*
Summer 2023	Theoretical Biological Physics	*Les Houches School of Physics*

SERVICE & OTHER WORK

2025–2026 **"LS50: Integrated Science", Harvard University** · *Instructor* — *Cambridge, MA*
Designed a lab class where student groups will tackle a novel research problem—exploring HIV bnAb evolution with high-throughput antibody libraries and statistical analysis. At the end of the academic year, we selected a group of undergraduates to write a short paper for submission to an academic journal (currently in preparation).

2024–2025 **Tutoring Plus** · *Volunteer tutor* — *Cambridge, MA*
Volunteered with local non-profit to provide free tutoring services to K-12 students.

2022 **"Data Science for Biology", UC Berkeley** · *Graduate Student Instructor* — *Berkeley, CA*
Prepared and gave lectures for class discussion section. Held office hours for students. Helped to design coding assignments and graded them.

2017 **"Biocomputing", Universitat Pompeu Fabra** · *Teaching Assistant* — *Barcelona, Spain*
Helped second year Bioengineering undergraduate students with coding assignments (in Spanish). Prepared and gave lectures for class discussion section.

2011–2019 **Virginia Science Olympiad** · *Volunteer* *Northern Virginia and remotely*
Writer for regional and state Science Olympiad forensics events for Middle School students. Continue to work remotely since moving away from Virginia.

2016–2017 **Fulbright Outreach** · *Volunteer* *Barcelona, Spain*
Gave talks to high school students in Barcelona about topics in modern biology research, and my own research projects.

2016–2017 **Brown College, Rice University** · *Academic Fellow* *Houston, TX*
Tutored individual college members in a variety of subjects, including physics, math and computational engineering. Led review sessions before exams for freshman physics & math classes.

2014–2016 **Biomedical Engineering Society** · *Mentor* *Houston, TX*
Served as a year-long resource and mentor for underclassman bioengineers. Advised mentees on class selection, internships, research, etc.

2014 **"Bioengineering Fundamentals", Rice University** · *Teaching Assistant & Grader* *Houston, TX*
Graded homework and led homework sessions for introductory bioengineering class.